Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	Membrane-Bound Erythropoietin–Expressing Cultured Red Blood Cells for the Effective Anemia Therapy			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI ( ), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Cell and Tissue Therapeutics	Medical Bio	Drug Delivery Systems
Concept of Proposed Technology	The proposed technology aims to develop a next-generation cell-based therapeutic platform: Membrane-Bound Erythropoietin-expressing cultured Red Blood Cells (MB-EPO cRBCs) for the effective anemia therapy. **The Challenge:** Chronic anemia, whether resulting from renal failure, myelosuppression (cancer/infection), chronic inflammation (elevated hepcidin), or the physiological rigors of spaceflight, remains a significant burden on quality of life. Current standards of care, such as recombinant EPO injections and donor transfusions, are often hampered by transient efficacy, iron overload, and systemic side effects. **The Solution:** This platform leverages engineered stem cell derivatives, $CD34^+$ hematopoietic stem/progenitor cells (HSPCs), from cord blood (CB), human induced pluripotent stem cells (hiPSCs), or matched donors, to create a renewable source of therapeutic cells. By genetically modifying these HSPCs, we ensure that all progeny (the cRBCs) consistently express the MB-EPO construct.			

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	MB-EPO cRBCs are designed to perform dual therapeutic functions: **1. Oxygenation:** Direct supplementation of oxygen-carrying capacity via the transfused erythroid mass. **2. Niche-Targeted Signaling:** As these cells circulate and eventually home to the splenic or bone marrow microenvironments during their natural life cycle, the membrane-bound EPO provides localized, contact-dependent stimulation of endogenous erythropoiesis. **Key Innovation:** Unlike systemic EPO, which circulates freely and can cause off-target effects, MB-EPO remains tethered to the cell surface via a specialized molecular linker, ensuring signaling is restricted to the erythroid niche. By providing sustained, localized EPO signaling, this approach is intended to reduce dependence on repeated systemic EPO administration while supporting physiological red blood cell production. The platform is based on **engineered erythroid progenitor cells** derived from pluripotent stem cell sources or immortalized progenitors. While the primary focus is anemia treatment, the underlying technology is potentially adaptable to additional hematologic and systemic conditions following further validation.
Rationale for Joint R&D Project	The Development of functional therapeutic cRBCs from HSPCs requires a multidisciplinary approach. A joint R&D framework is essential to manage the following work packages: • **Advanced Cell Engineering:** Optimizing the genetic construct within HSPCs (from hiPSCs, CB, or donors) to ensure stable MB-EPO expression throughout the differentiation process. • **Bioprocess Scaling:** Transitioning from static culture to high-density stirred-tank bioreactors capable of producing cell volumes necessary for clinical administration. • **Linker Kinetics:** Utilizing protein engineering to refine the presentation of EPO, ensuring the anchoring mechanism is durable enough for the intended therapeutic window. • **Translational Modeling:** Testing stem-cell-derived products in specialized anemia disease models to evaluate *in vivo* safety and efficacy. Development of functional therapeutic cultured red blood cells requires advanced cell engineering, scalable manufacturing, and rigorous *in vitro* and *in vivo* validation**,** which cannot be achieved by a single organization alone. The joint R&D framework enables shared responsibility for: • Proof-of-concept (PoC) studies and optimization of membrane-bound EPO expression • Mechanistic study (MOA) of EPO from MB-EPO cRBC • Verification and refinement of EPO release kinetics through linker modification • Development of scalable MB-EPO cRBC production processes and bioreactor systems • Establishment of relevant anemia disease animal models and comprehensive *in vivo* evaluation International collaboration allows the integration of expertise in stem cell biology, synthetic biology, and biomanufacturing to accelerate technology maturation toward translational readiness.
Key Words	• Engineered cultured red blood cells • Membrane-bound erythropoietin • Cell-based therapy for anemia • Pluripotent stem cell–derived erythroid progenitors • Renal anemia and spaceflight-associated anemia

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (4)
R&D Objectives	• Establishing MB-EPO cRBC production processes and CMC-related foundations • Generating immortalized erythroid progenitor cell lines expressing membrane-bound EPO • Optimizing erythroid differentiation and enucleation efficiency • Verification of MOA *via* systemic mechanism study of MB-EPO cRBC • *In vitro* evaluation and analysis • Developing relevant anemia disease animal models, including humanized mouse models • Characterizing EPO release pharmacokinetics and pharmacodynamics *via* linker modification • Developing and validating bioreactor systems for scale-up production • Evaluating *in vivo* efficacy, safety, and lifespan of MB-EPO cRBC			
R&D Target Specifications	• **Cell proliferation rate:** ≥ 100-fold expansion • **Cell differentiation rate:** ≥ 70% • **RBC enucleation rate:** ≥ 45% • **EPO half-life:** ≥ 1 month • **In vivo safety:** No observed toxicity • **In vivo efficacy:** Restoration of hemoglobin levels comparable to normal ranges			
R&D Plan	*Year 1* **Key Tasks** • Establishing erythroid differentiation conditions based on human pluripotent stem cells • Generating EPO knock-in inducible pluripotent stem cells using gene-editing technology • Characterizing inducible EPO expression mechanisms **Milestones** • Preparation of donor-derived pluripotent stem cell lines (*via* IRB approval) • Establishment of EPO knock-in progenitor cell lines • Verification of inducible EPO expression control *Year 2* **Key Tasks** • Immortalization of EPO-expressing erythroid progenitor cell lines • Optimizing proliferation and differentiation conditions in suspension culture • Improving enucleation efficiency through bone marrow microenvironment simulation • Conducting *in vitro* evaluation and mechanism study **Milestones** • Establishment of research cell banking • Development of bioreactor-based manufacturing processes			

	• Improved enucleation rate and product purity • Verification of MOA *Year 3* **Key Tasks** • Developing standardized evaluation and reporting guidelines • Conducting *in vitro* efficacy assessments • Establishing anemia disease mouse models • performing *in vivo* efficacy and safety studies **Milestones** • Standardized test report guidelines established • Completion of *in vitro* efficacy and viability evaluation • Completion of *in vivo* efficacy and safety evaluation

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• Confirming inducible EPO expression mechanisms (MOA) • Supporting scale-up strategies for erythroid progenitor production • Optimizing EPO expression systems and linker designs • Developing immortalization strategies and validate genetic stability • Optimizing differentiation, proliferation, and enucleation conditions • Developing evaluation methods and standardized testing guidelines • Conducting *in vitro* efficacy studies
Expected R&R of Korean SME Partner	• Establishing differentiation protocols using pluripotent stem cell lines • Obtaining ethical approvals and donor material for cell sourcing • Advancing differentiation, proliferation and enucleation technologies • Establishing bioreactor-based suspension culture systems • Advancing isolation methods and scale-up production process • Developing analytical methods and quality evaluation criteria • Generating pilot data and preliminary test reports • Conducting *in vivo* viability, safety, and efficacy assessments

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	Expected Outcomes • Establishment of a functional MB-EPO cRBC platform for anemia treatment • Generation of scalable, standardized production and evaluation processes • Demonstration of *in vivo* efficacy and safety in relevant anemia models Scientific and Societal Impact • Advancement of regenerative medicine and artificial blood research • Development of novel treatment options for chronic and severe anemia • Development of combination therapy options for cancer therapy or infection treatment or anti-aging/chronic disease therapy that result in significant anemia • Strengthening of international collaboration in cell-based therapeutics

6. References

(1) Renal Anemia: Anemia in Chronic Kidney Disease, https://www.niddk.nih.gov/health-information/kidney-disease/anemia

(2) Space Anemia: Investigating anemia in space, https://www.asc-csa.gc.ca/eng/sciences/spa2.asp; Hemolysis contributes to anemia during long-duration space flight, Guy Trudel et. al. *Nature Medicine* volume 28, pages59–62 (2022) https://www.nature.com/articles/s41591-021-01637-7